

What the visit counts remember

Working note for direction S2 (independent derivation, not yet compared with the S2 notes)

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1 The question and the answer

A chain on d states switches slowly. We do not see its path. We only see how long it spent in each state (the count vector of Paper C, divided by N). Which rate matrices can be told apart from this, and how much is lost compared with seeing the path?

Here is what we find. The proofs are in the sections that follow.

1. *One horizon is every horizon* (Theorem 3.1). The law of the counts at a single time fixes their law at every time.
2. *The counts know each pair of flows, but not which is which* (Theorem 4.1). For two states i and j , the counts fix the flow from i to j and the flow from j to i as an unordered pair. The exit rates then decide the order for almost every chain, so almost every chain can be recovered (Corollary 4.3).
3. *A stationary start hides the arrow of time, and nothing else* (Proposition 2.1, Theorem 5.2). A chain and its reversal give the same counts. These are the only two candidates whenever the rate matrix has no cut (no way to split the states into two groups of two or more that talk to each other through rank one blocks), and almost every chain is like this. The size of the net flow on every pair is still known, so the counts say how far the chain is from reversible but not which way it turns.
4. *Lumping gives more ties* (Theorem 6.1). If a group of states looks like a single state from outside, we can reverse the outside and keep the inside, or reverse the inside and keep the outside. This gives two chains on four states, with every rate positive and a start that is not stationary, that no amount of count data can tell apart.
5. *What is lost* (Theorems 7.1 and 7.2). When switches are rare, the counts are as good as the full path for the two-way flow on each pair, and blind to first order to the net flow. With a stationary start the information about the net flow vanishes exactly at reversible chains, at every horizon.

2 Setting

Let $G = (g_{ij})$ be a rate matrix on $[d] = \{1, \dots, d\}$, so $g_{ij} \geq 0$ for $i \neq j$ and $q_i = -g_{ii} = \sum_{j \neq i} g_{ij}$ is the exit rate of i . Let Y be the chain with rates G and start law μ , and let

$$L_i(t) = \int_0^t \mathbf{1}\{Y_s = i\} ds, \quad L(t) = (L_1(t), \dots, L_d(t)).$$

This is the limit of Paper C's setting. If X is the N -step chain with transition matrix $I + G/N$, then its count vector K_N satisfies $K_N/N \rightarrow L(1)$ in law. With $G = TQ$ the number T counts the switches, as in Papers A to C.

We use three numbers for each pair of states. The *flows* are $f_{ij} = \mu_i g_{ij}$ (how much mass moves from i to j at time 0). The *two-way flow* is $c_{ij} = f_{ij} + f_{ji}$ and the *net flow* is $J_{ij} = f_{ij} - f_{ji}$. We also write $a_{ij} = g_{ij} g_{ji}$, the weight of a round trip. Note that

$$f_{ij} + f_{ji} = c_{ij}, \quad f_{ij} f_{ji} = \mu_i \mu_j a_{ij}, \quad J_{ij}^2 = c_{ij}^2 - 4\mu_i \mu_j a_{ij}, \quad (1)$$

and that $\sum_j J_{ij} = -(\mu G)_i$. So μ is stationary exactly when the net flow obeys Kirchhoff's law at every state, and the chain is reversible with respect to μ exactly when $J = 0$. (In the physics literature c_{ij} is the traffic or dynamical activity of the pair and J_{ij} is its current, and the times L are the standard example of an observable that does not change when time is reversed [10].)

Write $\Lambda = \text{diag}(\lambda)$. By the Feynman-Kac formula, for $\lambda \in \mathbb{C}^d$,

$$\Phi_t(\lambda) = \mathbb{E} e^{-\lambda \cdot L(t)} = \mu^\top e^{t(G-\Lambda)} \mathbf{1}, \quad r(\lambda) = \int_0^\infty \Phi_t(\lambda) dt = \mu^\top (\Lambda - G)^{-1} \mathbf{1}, \quad (2)$$

the second for λ with positive coordinates. The function r is rational, which is what makes it easy to read.

Proposition 2.1 (reversal). *Let $\mu = \pi$ be stationary for G with $\pi_i > 0$, and let $\hat{g}_{ij} = \pi_j g_{ji} / \pi_i$. Then $(\pi, \hat{G})$ and (π, G) give the same law of $L(t)$ for every t .*

Proof. Run the film backward. The path $s \mapsto Y_{t-s}$ is the stationary chain with rates $\hat{G}$, and it spends the same time in each state as Y does. $\square$

In terms of flows, reversal swaps f_{ij} and f_{ji} on every pair. So it keeps c and flips the sign of J .

3 One horizon is every horizon

Theorem 3.1. *Let G be irreducible. Suppose a second pair (μ', G') gives the same law of $L(1)$. Then it gives the same law of $L(t)$ for every $t > 0$, and $r' = r$.*

The idea is that each state contributes one mode $e^{z_i(\lambda)}$ to Φ_1 , where z_i is an eigenvalue of $G - \Lambda$. Exponentials of different algebraic functions cannot cancel each other, so each mode can be read off from Φ_1 . The modes are the poles of the resolvent, and the resolvent knows every horizon. We need two lemmas. Both are known. The first is a weak case of theorems on the independence of exponentials [3], and the second is Lemma 5.1 of [2]. We include short proofs.

Lemma 3.2 (exponentials do not cancel). *Let $B \subset \mathbb{C}^d$ be a ball and let $w_1, \dots, w_m$ and $c_1, \dots, c_m$ be holomorphic functions on B that are algebraic over $\mathbb{C}(\lambda)$. Suppose $w_a - w_b$ is not constant for $a \neq b$. If $\sum_a c_a e^{w_a} = 0$ on B , then every c_a is zero.*

Proof. We use induction on m . The case $m = 1$ is clear. Suppose the claim holds for $m - 1$ and that $c_m \neq 0$ (otherwise we have fewer terms). On a smaller ball where c_m has no zero, divide by $c_m e^{w_m}$ and differentiate in λ_j . With $b_a = c_a / c_m$ and $v_a = w_a - w_m$ this gives $\sum_{a < m} (\partial_j b_a + b_a \partial_j v_a) e^{v_a} = 0$, which has $m - 1$ terms. By induction $\partial_j (b_a e^{v_a}) = 0$ for every j and every $a < m$, so $b_a e^{v_a}$ is a constant k_a . If some $k_a \neq 0$, then $e^{v_a} = k_a / b_a$ is algebraic. This is impossible. Restrict to a complex line on which v_a is not constant. There v_a is a nonconstant meromorphic function on a compact Riemann surface, so it has a pole, and near the pole e^{v_a} has an essential singularity, which an algebraic function does not have. Hence every $k_a = 0$, so $c_a = 0$ for $a < m$ (on the smaller ball, and so on B), and then $c_m = 0$, which contradicts our assumption. $\square$

Lemma 3.3 (one polynomial). *If G is irreducible, then $\chi(z, \lambda) = \det(z + \Lambda - G)$ is an irreducible polynomial.*

Proof. Put $x_i = z + \lambda_i + q_i$ and let $A \geq 0$ be G without its diagonal. Then $\chi = \det(X - A) = \sum_{U \subseteq [d]} m(U) \prod_{i \notin U} x_i$ with $m(U) = \det(-A_{UU})$. This has degree one in each x_i . So if $\chi = pq$ with both factors nonconstant, the variables split into two nonempty sets S and S^c , with p using only S and q only S^c , and comparing coefficients gives $m(U \cup V) = m(U)m(V)$ for $U \subseteq S$ and $V \subseteq S^c$.

Since G is irreducible, some cycle of positive rates meets both S and S^c . Take one with the fewest states, let W be its set of states, and put $U = W \cap S$, $V = W \cap S^c$. Expand $m(W)$ over permutations of W . The permutations that keep U and V apart sum to $m(U)m(V)$. Any other permutation with nonzero weight contains a cycle that meets both sides, which by our choice uses all of W . A single cycle on W has sign $(-1)^{|W|-1}$ and weight $(-1)^{|W|}$ times a product of rates, so it contributes minus that product. These contributions cannot cancel, and ours is not zero. So $m(W) \neq m(U)m(V)$, a contradiction. $\square$

Proof of Theorem 3.1. Both Φ_1 and Φ'_1 are entire and agree on $\mathbb{R}^d$, so they agree on $\mathbb{C}^d$. Let $m = 2 \max_i \max(q_i, q'_i)$ and let B be the ball of radius $3m$ around the real point $(0, 12m, 24m, \dots)$. On B we have $|\lambda_i - \lambda_j| > 6m$ for $i \neq j$. By Gershgorin's theorem, the matrix $G - \Lambda$ then has exactly one eigenvalue $z_i(\lambda)$ in the disc $|z + \lambda_i + q_i| \leq q_i$ for each i , and these discs are disjoint. So the z_i are simple, holomorphic and algebraic on B , and

$$\Phi_1 = \sum_i \rho_i e^{z_i}, \quad \rho_i = \frac{n(z_i, \lambda)}{\partial_z \chi(z_i, \lambda)}, \quad n(z, \lambda) = \mu^\top \text{adj}(z + \Lambda - G) \mathbf{1},$$

where ρ_i is the residue of $n/\chi = \mu^\top (z + \Lambda - G)^{-1} \mathbf{1}$ at z_i . Since the residues of a resolvent sum to the identity, $\sum_i \rho_i = 1$. The same holds for (μ', G') with z'_i , ρ'_i and χ' .

No mode is missing. If $\rho_i = 0$ on B , then n and χ share the root z_i , so they have a common factor. But χ is irreducible (Lemma 3.3), and n has degree $d - 1$ in z with leading coefficient $\sum_i \mu_i = 1$.

The modes match. Since $|z_i + \lambda_i| \leq m$ and $|z'_j + \lambda_j| \leq m$ on B , while $\lambda_i - \lambda_j$ moves by more than $4m$ across B , none of the differences $z_i - z_j$, $z'_i - z'_j$ and $z_i - z'_j$ with $i \neq j$ is constant. Apply Lemma 3.2 to $\sum_i \rho_i e^{z_i} - \sum_i \rho'_i e^{z'_i} = 0$, after merging any two exponents that differ by a constant. The term $\rho_i e^{z_i}$ cannot stand alone, so $z'_i = z_i - \kappa_i$ for a constant κ_i , and $\rho'_i = e^{\kappa_i} \rho_i$.

The constants vanish. The polynomials $\chi'(z, \lambda)$ and $\chi(z + \kappa_1, \lambda)$ share the root $z_1 - \kappa_1$, have the same degree and leading term, and the second is irreducible. So they are equal. Comparing roots in the disjoint discs gives $\kappa_i = \kappa_1$ for every i . Then $1 = \sum_i \rho'_i = e^{\kappa_1} \sum_i \rho_i = e^{\kappa_1}$, and κ_1 is real because the eigenvalues at the center of B are real. Hence every $\kappa_i = 0$.

So $\chi' = \chi$ and $\rho'_i = \rho_i$, which gives $\mu'^\top (z + \Lambda - G')^{-1} \mathbf{1} = \sum_i \rho_i / (z - z_i) = \mu^\top (z + \Lambda - G)^{-1} \mathbf{1}$ on B , and so everywhere. The left and right sides are the Laplace transforms in t of Φ'_t and Φ_t . $\square$

4 What the counts know about a pair

Theorem 4.1. *Let G be irreducible and $\mu_i > 0$ for every i . The law of $L(1)$ determines μ_i and q_i for every state, and a_{ij} and c_{ij} for every pair. So by (1) it determines $\{f_{ij}, f_{ji}\}$ as an unordered pair, namely the two roots of $x^2 - c_{ij}x + \mu_i \mu_j a_{ij}$, and with it $|J_{ij}|$.*

Proof. By Theorem 3.1 the law determines r . For a set F of states let $\lambda_k \rightarrow \infty$ for every $k \notin F$. Then $e^{-\lambda \cdot L(t)}$ decreases to $e^{-\lambda \cdot L(t)} \mathbf{1}\{Y \text{ stays in } F \text{ up to } t\}$, so r tends to $r_F = \mu_F^\top (\Lambda_F - G_{FF})^{-1} \mathbf{1}$,

the same function for the chain killed when it leaves F . With $x_i = \lambda_i + q_i$,

$$r_{\{i\}} = \frac{\mu_i}{x_i}, \quad r_{\{i,j\}} = \frac{\mu_i x_j + \mu_j x_i + c_{ij}}{x_i x_j - a_{ij}}.$$

The first gives μ_i and q_i . In the second, cross-multiplying two such fractions and comparing the coefficients of $x_i x_j$ and then of x_i shows that c_{ij} and a_{ij} are determined. $\square$

The formula for $r_{\{i,j\}}$ says what is going on. A path that jumps once between i and j leaves the same counts whether it went from i to j at time s or from j to i at time $1 - s$. So one-jump paths show only the sum c_{ij} . Round trips show the product a_{ij} . A sum and a product give two numbers, but not their order.

Let S be a set of pairs. Write G^S for the matrix obtained from G by swapping f_{ij} and f_{ji} on every pair in S (so $g_{ij}^S = \mu_j g_{ji}/\mu_i$ there), with the diagonal chosen so that rows sum to zero.

Corollary 4.2 (the exit rates decide). *Under the hypotheses of Theorem 4.1, every (μ', G') with the same law of $L(1)$ has $\mu' = \mu$ and $G' = G^S$ for a set S of pairs on which the net flow obeys Kirchhoff's law,*

$$\sum_{j:\{i,j\} \in S} J_{ij} = 0 \quad \text{for every state } i. \quad (3)$$

Proof. By Theorem 4.1, G' has the same μ , the same exit rates and the same unordered pairs of flows. Let S be the set of pairs where the order differs. The exit rate of i is $\sum_j f_{ij}/\mu_i$, and swapping changes it by $-\sum_{j:\{i,j\} \in S} J_{ij}/\mu_i$. $\square$

Corollary 4.3. *Let G be irreducible and $\mu_i > 0$ for every i .*

- (a) *If $d = 2$, the law of $L(1)$ determines (μ, G) .*
- (b) *If $d = 3$, it determines (μ, G) unless μ is stationary and the chain is not reversible. In that case exactly two pairs have this law, namely (μ, G) and its reversal.*
- (c) *For every μ and almost every G , it determines (μ, G) .*
- (d) *If μ is the stationary law, then for almost every G exactly two pairs have this law, namely (μ, G) and its reversal.*

Proof. Swapping a pair with $J_{ij} = 0$ changes nothing, so we may take $J_{ij} \neq 0$ on S . In (a) a single pair cannot satisfy (3). In (b), a state that lies in exactly one pair of S violates (3), so S is all three pairs and $J_{12} = J_{23} = J_{31}$. Then $\sum_j J_{ij} = 0$ at every state, so μ is stationary, G^S is the reversal, and Proposition 2.1 shows that it does have the same law. In (c), fix μ and use the flows as coordinates. For each nonempty S , condition (3) at a state that meets S is a linear equation that is not trivial, so it fails off a hyperplane. In (d), use the coordinates (π, c, J) , where J runs over the flows that obey Kirchhoff's law on the whole graph. Let S be neither empty nor everything. Some state i has a pair $\{i, j\}$ in S and a pair $\{i, k\}$ outside it, and the unit flow around the triangle $i \rightarrow j \rightarrow k \rightarrow i$ violates (3) at i . So again (3) fails off a proper subspace. What is left is S empty or everything, which gives G and its reversal. $\square$

Remark 4.4. The proof is also a recipe. Read μ_i and q_i from the paths that never move, and c_{ij} and a_{ij} from the paths that visit two states. Then $f_{ij} = \frac{1}{2}(c_{ij} \pm |J_{ij}|)$, and the signs are the ones for which $\sum_j f_{ij} = \mu_i q_i$ at every state. So paths that visit at most two states are enough for almost every chain.

Remark 4.5. Part (d) is sharper than “the counts cannot see irreversibility”. They cannot see its direction. They do see $|J_{ij}|$ on every pair, so they know exactly how far the chain is from reversible. For long horizons it is known that the variance of L and its large deviations depend on the net flow only through even expressions [5, 8]. Here the statement is exact, at every horizon, and for the whole law.

5 Chains without a cut

Theorem 4.1 reads the law through the paths that visit at most two states. Reading all of it leads to an old question of linear algebra, namely when two matrices have the same principal minors.

Proposition 5.1 (two sets of minors). *Let G be irreducible. The law of $L(1)$ determines every principal minor of G and every principal minor of $G - \mathbf{1}\mu^\top$. Conversely, these minors determine the law of $L(t)$ for every t .*

Proof. By Theorem 3.1 the law determines $r = n_0/\chi_0$, where $\chi_0(\lambda) = \det(\Lambda - G)$ and $n_0(\lambda) = \mu^\top \text{adj}(\Lambda - G)\mathbf{1}$. The polynomial χ_0 is irreducible (the proof of Lemma 3.3 with $z = 0$), the degree of n_0 is smaller, and the coefficient of $\lambda_1 \cdots \lambda_d$ in χ_0 is 1. So the fraction is in lowest terms, and r determines χ_0 and n_0 . For a matrix A we have $\det(\Lambda - A) = \sum_U \det(-A_{UU}) \prod_{i \notin U} \lambda_i$, so the coefficients of χ_0 are the principal minors of $-G$. The same holds for $\chi_0 + n_0$ and $-(G - \mathbf{1}\mu^\top)$, because $\det(\Lambda - G + \mathbf{1}\mu^\top) = \chi_0 + n_0$ (the determinant of a rank one change). For the converse, the minors give r , and $r(\lambda + z\mathbf{1})$ is the Laplace transform of $t \mapsto \Phi_t(\lambda)$ since the coordinates of $L(t)$ sum to t . $\square$

Two things always keep the principal minors of a matrix A , namely passing to DAD^{-1} for a diagonal matrix D and passing to the transpose. For a rate matrix these are familiar. The transpose with the right D is the reversal, since $\hat{G} = \Pi^{-1}G^\top\Pi$ with $\Pi = \text{diag}(\pi)$. Hartfiel and Loewy [7] and Loewy [9] found when nothing else keeps them. Say that G has a *cut* if there is a set X of states with $2 \leq |X| \leq d - 2$ such that both blocks G_{XX^c} and G_{X^cX} have rank at most one. (So a chain on two or three states has no cut.) Their theorem says that if A is irreducible and has no cut, then every matrix with the same principal minors as A is DAD^{-1} or $DA^\top D^{-1}$ for some diagonal D (see also [2, Theorem 4.1]).

Theorem 5.2 (at most two candidates). *Let G be irreducible with no cut, and let π be its stationary law. Suppose (μ', G') gives the same law of $L(1)$ as (μ, G) . Then $\mu' = \mu$, and G' is either G or its reversal $\hat{G}$. Moreover $(\mu, \hat{G})$ does give the same law if and only if, for the stationary chain with rates G and $w_i = \mu_i/\pi_i$,*

$$\mathbb{E}_\pi[w(Y_0) \mid L(1)] = \mathbb{E}_\pi[w(Y_1) \mid L(1)]. \quad (4)$$

In words, a chain with no cut has at most one rival, its own reversal, and the rival passes exactly when the counts cannot tell the first state from the last as far as the weight w is concerned. A stationary start has $w = 1$, and then (4) is clear.

Proof. By Proposition 5.1, G' has the same principal minors as G . By the theorem of Hartfiel and Loewy, $G' = DGD^{-1}$ or $G' = DG^\top D^{-1}$. We use that the rows of G' sum to zero. In the first case $GD^{-1}\mathbf{1} = 0$. Since G is irreducible, its kernel is spanned by $\mathbf{1}$, so D is a multiple of the identity and $G' = G$. In the second case $G^\top D^{-1}\mathbf{1} = 0$, so $D^{-1}\mathbf{1}$ is a multiple of π and $G' = \Pi^{-1}G^\top\Pi = \hat{G}$. Next, the coefficient of $\prod_{k \neq i} \lambda_k$ in n_0 is μ_i , so $\mu' = \mu$.

For the last claim, start the chain with rates $\hat{G}$ from μ . Its law is the stationary law of that chain weighted by $w(Y_0)$. Run the film backward, as in Proposition 2.1. This turns it into the stationary chain with rates G weighted by $w(Y_1)$, with the same $L(1)$. On the other hand (μ, G) is the stationary chain with rates G weighted by $w(Y_0)$. So the two laws of $L(1)$ agree exactly when $w(Y_0) - w(Y_1)$ has conditional mean zero given $L(1)$. $\square$

A cut asks for 2×2 minors to vanish, so almost every rate matrix has none. Thus Theorem 5.2 names the exceptional chains in parts (c) and (d) of Corollary 4.3. They all have a cut, or (in part (c)) they satisfy (4) with a start that is not stationary. The second can happen, as we will see after Theorem 6.1. For a chain with every rate drawn at random it does not happen (`check_rigidity.py`).

The same two operations, and the same theorem of Loewy, appear for the kernels of permanental processes, which describe local times of Markov processes at certain random times [11]. What is different here is the fixed horizon and the start law, which bring in the second matrix $G - \mathbf{1}\mu^\top$.

6 Lumping gives more ties

A chain with a cut can have other rivals, and a start that is not stationary does not save us. The reason is that a chain can contain a smaller chain to which Proposition 2.1 applies.

Call a set M of states a *module* if, with O its complement,

- (i) every state of M leaves to $x \in O$ at the same rate, $g_{bx} = \gamma_x$,
- (ii) every entrance into M lands the same way, $g_{xb} = r_x \nu_b$ for a law ν on M ,
- (iii) the start inside M follows ν , namely $\mu_b = \mu(M) \nu_b$.

From outside, M then looks like one state $*$. The *lumped chain* lives on $O \cup \{*\}$ with rates g_{xy} inside O , rates r_x into $*$ and γ_x out of it, and start law $\bar{\mu}$ (with $\bar{\mu}_* = \mu(M)$). The *inside chain* lives on M with the rates $g_{bb'}$. By (i) and (ii) the blocks G_{MO} and G_{OM} have rank one, so a module with two or more states on each side is a cut.

Theorem 6.1 (two reversals). *Let M be a module.*

- (a) *(Reverse the outside, keep the inside.) Suppose $\bar{\mu}$ is stationary for the lumped chain. Reverse the lumped chain with respect to $\bar{\mu}$, and let G' be the chain with the reversed rates on O , the reversed rates $\hat{r}_x \nu_b$ into M and $\hat{\gamma}_x$ out of M , and the old rates $g_{bb'}$ inside M . Then (μ, G') and (μ, G) give the same law of $L(t)$ for every t .*
- (b) *(Reverse the inside, keep the outside.) Suppose ν is stationary for the inside chain. Let G'' be G with the inside chain replaced by its reversal with respect to ν . Then (μ, G'') and (μ, G) give the same law of $L(t)$ for every t .*

Proof. Build Y in two steps. First run the lumped chain Z . Then, for each visit of Z to $*$, fill the visit with a fresh copy of the inside chain (rates $g_{bb'}$ on M , started from ν), run for the length of the visit. This is the chain (μ, G) . By (i) the time and place of an exit do not depend on what happens inside M , and by (ii) and (iii) every visit starts from ν .

So the counts are a function of two things, the times Z spends in the states of O , and the list of lengths of its visits to $*$, together with independent fillings whose law depends only on those lengths and not on their order.

(a) Run the film of Z backward. Since $\bar{\mu}$ is stationary, this is the reversed lumped chain. It has the same times in O and the same list of visit lengths. Filling its visits in the same way builds the chain (μ, G') .

(b) Keep Z and run the film of each filling backward. A filling is the inside chain started from its stationary law ν and run for a fixed time, so its film run backward is the reversed inside chain started from ν , with the same times in the states of M . This builds the chain (μ, G'') . $\square$

A single state is always a module, and then (a) is Proposition 2.1. Part (b) asks nothing of the start outside M or of the lumped chain. On the level of the polynomial χ_0 these ties are known. Every matrix with the same principal minors as an irreducible matrix is reached from it by diagonal similarity, the transpose, and transposing one side of a cut [7, 2, 4], and (a) and (b) are of this kind. What Theorem 6.1 adds is that they pass the second test of Proposition 5.1, the one that involves the start.

Example 6.2 (the reversal can win without a stationary start). Let M have three states and O a single state z , with every rate positive, and let ν be stationary for the inside chain. Here M is not a cut, since $|O| = 1$, and for a typical choice of the rates G has no cut at all. Balance at z gives $\pi_z g_{zb} = \pi_b \gamma_z$, so π is proportional to ν on M , and one checks that $G'' = \hat{G}$. By (b), the start $\mu = a\nu + (1-a)\delta_z$ gives the same counts under G and under $\hat{G}$ for every a , and it is stationary for only one value of a . So (4) can hold with w not constant. Here w takes one value on M and another at z .

Example 6.3 (four states). Take $O = \{1, 2\}$ and $M = \{3, 4\}$. Let the lumped chain on $\{1, 2, *\}$ have rates $g_{12} = 0.9$, $g_{21} = 0.3$, $r_1 = 0.5$, $r_2 = 1.1$, $\gamma_1 = 0.8$, $\gamma_2 = 0.4$, and let $\bar{\mu}$ be its stationary law. Take $\nu = (0.35, 0.65)$ and inside rates $g_{34} = 1.7$, $g_{43} = 0.2$. Every rate of G is positive, and μ is not stationary for G , because ν is not stationary for the inside. The lumped chain is not reversible, so $G' \neq G$ (their entries differ by up to 0.59). The two laws of $L(t)$ agree to 10^{-15} at $t = 0.3, 1$ and 4 . In the language of Corollary 4.2, S is the five pairs that are not inside M .

When both parts apply, and neither the inside chain nor the lumped chain is reversible, we get four chains with the same counts, namely G , the chain with only the outside reversed, the chain with only the inside reversed, and the chain with both reversed (checked on six states with a stationary start, where the last one is $\hat{G}$).

7 What is lost

Now let $G = TQ$, with μ known and positive and every $Q_{ij} > 0$. We compare the Fisher information about Q in the full path on $[0, 1]$, written I_{path} , with the information in $L(1)$, written I_L . Here $c_{ij} = \mu_i Q_{ij} + \mu_j Q_{ji}$ and $q_i = \sum_j Q_{ij}$ are functions of Q , and for three states we put

$$\tau_{ijk} = \sum_{(x,y,z)} \mu_x Q_{xy} Q_{yz} = \sum_y \frac{c_{xy} c_{yz} + J_{xy} J_{yz}}{2\mu_y},$$

where the first sum is over the six orders of $\{i, j, k\}$ and in the second x and z are the two states other than y .

Theorem 7.1 (rare switches). *As $T \rightarrow 0$,*

$$I_{\text{path}} = T \operatorname{diag}\left(\frac{\mu_i}{Q_{ij}}\right) + O(T^2), \quad I_L = T \sum_{i < j} \frac{\nabla c_{ij} \nabla c_{ij}^\top}{c_{ij}} + O(T^2).$$

So the efficiencies (the eigenvalues of $I_{\text{path}}^{-1}I_L$) split in half, with $\binom{d}{2}$ of them tending to 1 and $\binom{d}{2}$ tending to 0. The counts are first-order efficient for every two-way flow c_{ij} and carry no first-order information about any net flow. For a direction v with $v \cdot \nabla c_{ij} = 0$ for all pairs,

$$v^\top I_L v = T^2 \left[\sum_i \mu_i (v \cdot \nabla q_i)^2 + \sum_{i < j < k} \frac{(v \cdot \nabla \tau_{ijk})^2}{2\tau_{ijk}} \right] + O(T^3). \quad (5)$$

Proof. The score of the path for Q_{ij} is $s_{ij} = N_{ij}/Q_{ij} - TL_i$, where N_{ij} counts the jumps from i to j , and $I_L = \mathbb{E}[\bar{s}\bar{s}^\top]$ with $\bar{s} = \mathbb{E}[s \mid L]$. Since $\mathbb{E}N_{ij} = T\mu_i Q_{ij} + O(T^2)$, the first claim follows. For the second we sort by the face of the simplex that holds L . The chance of three or more jumps is $O(T^3)$, and so is the second moment of the score on that event.

A vertex. $L = e_i$ means no jump, so $\bar{s} = -T\nabla q_i$ there, with probability $\mu_i + O(T)$. This contributes $T^2\mu_i \nabla q_i \nabla q_i^\top$.

An edge. Given L on the edge of i and j , up to a relative error $O(T)$ that is uniform along the edge, the path made one jump, from i to j with probability f_{ij}/c_{ij} and from j to i otherwise. So $\bar{s}_{ij} = \mu_i/c_{ij} + O(T)$ and $\bar{s}_{ji} = \mu_j/c_{ij} + O(T)$, that is, $\bar{s} = \nabla c_{ij}/c_{ij} + O(T)$. The edge has probability $Tc_{ij} + O(T^2)$. This contributes $T\nabla c_{ij} \nabla c_{ij}^\top / c_{ij} + O(T^2)$, and $O(T^3)$ in a direction v with $v \cdot \nabla c_{ij} = 0$.

A triangle. Given L in the open face of i, j and k , up to a uniform relative error $O(T)$ the path made two jumps in one of six orders. Each order fits every point of the face, and the order (x, y, z) has probability $\mu_x Q_{xy} Q_{yz} / \tau_{ijk}$. Since τ_{ijk} is linear in each rate, $\bar{s} = \nabla \tau_{ijk} / \tau_{ijk} + O(T)$. The face has probability $T^2 \tau_{ijk} / 2 + O(T^3)$, because the two jump times fill half of the unit square. This contributes the last term of (5).

For the efficiencies, in the coordinates f_{ij} the matrix $I_{\text{path}}^{-1}I_L$ tends to a block for each pair, equal to $\text{diag}(f_{ij}, f_{ji})$ times the all-ones matrix over c_{ij} , with eigenvalues 1 and 0. $\square$

When switches are rare, a typical path that tells us anything has one jump. The counts then say which two states and when, but not which way. This is why they keep the two-way flows and lose the net flows. The net flows come back at the next order from two places in (5), the paths that never move (their number gives the exit rates) and the paths that visit three states.

Theorem 7.2 (stationary start). *Let the start be the stationary law, and use the coordinates (π, c, J) for Q , where J obeys Kirchhoff's law at every state. Then the law of $L(t)$ is an even function of J . So at every reversible chain, and for every t , the information matrix of $L(t)$ vanishes on the J directions, a space of dimension $\binom{d-1}{2}$.*

Proof. Reversal sends (π, c, J) to $(\pi, c, -J)$ and keeps the law (Proposition 2.1). The derivative of an even function vanishes at 0, so the score in every J direction is zero at $J = 0$. The J directions are the cycle space of the complete graph, of dimension $\binom{d}{2} - (d - 1)$. $\square$

Remark 7.3. Near a reversible chain the law depends on J through quadratic terms such as the one in τ_{ijk} . So from n copies of the counts one should expect to learn J^2 at the usual rate, hence $|J|$ only at rate $n^{-1/4}$ near $J = 0$, and never its sign. We have not proved this. (Rates of this kind are known for models whose information matrix loses rank by one [12], which is the case of three states. There the model is assumed identifiable, and ours is not, because of the sign.) For three states and $T = 2$ the smallest eigenvalue of the information matrix is $0.94j^2, 0.96j^2$ at $j = 0.04, 0.02$, where j is the flow around the triangle.

The crossings of Paper C sit at the other end, where T is of order $\log N$ and so is large. There the counts keep the stationary law and, as far as we can see, nothing else. The first half of this is a theorem. Let π be the stationary law of Q , let $Z = (\mathbf{1}\pi^\top - Q)^{-1} - \mathbf{1}\pi^\top$, and let $\Gamma = \text{diag}(\pi)Z + Z^\top \text{diag}(\pi)$, the asymptotic covariance of the occupation times (so $\text{Cov } L(1) = \Gamma/T + O(T^{-2})$ for the generator TQ).

Theorem 7.4 (long horizons). *Let Q be irreducible with every $Q_{ij} > 0$, let the start be any fixed law or the stationary law, and let $D = \text{diag}(\pi_i/Q_{ij})$. Write π' for the matrix of the derivatives $\partial\pi_k/\partial Q_{ij}$.*

- (a) $\pi' D^{-1} \pi'^\top = \Gamma$. In words, the Cramér–Rao bound of the full path for π is the variance of the occupation fractions.
- (b) As $T \rightarrow \infty$, $I_{\text{path}} = TD + O(1)$ and $I_L \succeq T \pi'^\top \Gamma^+ \pi' - O(1)$, where Γ^+ is the inverse of Γ on the vectors with sum zero. So $d - 1$ of the efficiencies tend to 1.

Proof. (a) Differentiating $\pi^\top Q = 0$ and using $-QZ = I - \mathbf{1}\pi^\top$ gives $d\pi^\top = \pi^\top (dQ)Z$. Raising Q_{ij} (and lowering Q_{ii}) gives $\partial\pi/\partial Q_{ij} = \pi_i(Z_j - Z_i)$. So for a vector η and $u = Z\eta$,

$$\eta^\top \pi' D^{-1} \pi'^\top \eta = \sum_{i \neq j} \frac{Q_{ij}}{\pi_i} \pi_i^2 (u_j - u_i)^2 = \sum_{i \neq j} \pi_i Q_{ij} (u_j - u_i)^2.$$

On the other hand $\eta^\top \Gamma \eta = 2 \sum_i \pi_i \eta_i u_i$, and $-Qu = \eta - (\pi \cdot \eta)\mathbf{1}$ with $\pi \cdot u = 0$. Hence $\eta^\top \Gamma \eta = -2 \sum_i \pi_i u_i (Qu)_i$, and expanding the squares above with $Q\mathbf{1} = 0$ and $\pi^\top Q = 0$ gives the same number.

(b) The first claim is as in Theorem 7.1, since the chain spends the fraction $\pi_i + O(1/T)$ of its time in state i . For the second, let m and V be the mean and the covariance of $L(1)$ under the generator TQ , and let s be the score of $L(1)$. Then $\mathbb{E}[(L - m)s^\top] = m'$ (differentiate $\mathbb{E}L = m$), so the covariance matrix of the pair (L, s) has the blocks V , m' and I_L , and it is positive semidefinite. Its Schur complement gives $I_L \succeq m'^\top V^+ m'$. Now $e^{tQ} - \mathbf{1}\pi^\top$ decays exponentially, locally uniformly in Q and together with its derivatives in Q . So $m = \pi + O(1/T)$, $m' = \pi' + O(1/T)$ and $V = \Gamma/T + O(T^{-2})$, which gives the bound. Finally the nonzero eigenvalues of $D^{-1} \pi'^\top \Gamma^+ \pi'$ are those of $\Gamma^+ \pi' D^{-1} \pi'^\top = \Gamma^+ \Gamma$ by (a), namely 1 with multiplicity $d - 1$. Every efficiency is at most 1, and the eigenvalues of $I_{\text{path}}^{-1} I_L$ can only go up when I_L does. $\square$

So over a long horizon the counts estimate the stationary law as well as the full path does. That the occupation fractions are an efficient estimate of π is classical (see [1] for the path likelihood and [6] for efficiency). Part (a) is the identity behind it. The other $(d - 1)^2$ efficiencies appear to decay like $1/T$ (Table 1), but we have not proved that they tend to 0.

8 Checks and open questions

Each statement above was checked numerically in the scripts next to this note.

- `check_identifiability.py`. Reversal agrees to 10^{-15} for $d = 3, 4, 5$. The map from G to the law has full rank at random points for $d = 2, 3, 4$ with a known start, a stationary start and a start at one state. At reversible chains with a stationary start the rank drops by exactly $\binom{d-1}{2}$ for $d = 3, 4, 5$. Block reversal at a cut state works when the return rates are equal and fails (by $2 \cdot 10^{-3}$) when they are not.

T	known start	stationary start
0.05	0.98, 0.97, 0.95, 0.07, 0.04, 0.004	1.00, 1.00, 0.98, 0.95, 0.92, 0.001
1	0.75, 0.72, 0.50, 0.43, 0.42, 0.012	0.92, 0.90, 0.67, 0.50, 0.42, 0.005
20	0.99, 0.97, 0.04, 0.02, 0.01, 10^{-5}	0.99, 0.98, 0.04, 0.02, 0.01, 10^{-5}

Table 1: The six efficiencies of the counts for one chain on three states (exact, from the N -step chain with N at least 240). For small T three are lost with a known start, as in Theorem 7.1, and only the flow around the triangle with a stationary start. For large T only two survive (the stationary law, Theorem 7.4). This last regime is the one of Paper C’s crossings, where T is of order $\log N$.

- `check_lumping_decoder.py`. Example 6.3. The recipe of the remark after Corollary 4.3 returns one generator for a random start and two for a stationary start, for $d = 3, 4, 5$. For the N -step chain on three states the map from P to the law of K_N has full rank once $N \geq 3$.
- `check_information.py`. The two leading terms of Theorem 7.1 (the first to within 0.017 at $T = 0.025$, the second within 2% at $T = 0.02$ and still converging), Theorem 7.2 and Table 1.
- `check_rigidity.py`. Part (b) of Theorem 6.1 and Example 6.2 (the laws agree to 10^{-15} for $d = 4, 5, 7$). In each tie both sets of minors of Proposition 5.1 agree. For a random chain with a start that is not stationary, the reversal has the same minors of G but not of $G - \mathbf{1}\mu^\top$, and the laws differ.

What we do not know.

1. Is every tie built from the two reversals of Theorem 6.1? For a chain with no cut the answer is in Theorem 5.2, up to deciding (4). For a chain with a cut, the results of [4] list every G' with the same principal minors as G , so what is left is to find which of them also pass the test with $G - \mathbf{1}\mu^\top$. For four states with every rate positive, one step across one cut is enough to reach every such G' [4, Lemma 3.1].
2. When does (4) hold? A stationary start and Example 6.2 are the cases we know.
3. Does Theorem 3.1 have a form for the N -step chain? The proof uses that exponentials of algebraic functions do not cancel, and powers of algebraic functions can.
4. The large T row of Table 1. By Theorem 7.4, $d - 1$ efficiencies tend to 1. We expect the rest to decay like $1/T$.
5. A start at one state. The rank check says almost every chain is recovered, and the pair formulas change because some μ_i vanish. (Proposition 5.1 and Theorem 5.2 do not need $\mu_i > 0$, except that w must make sense.)

What is in the literature. A search (not a full review) found the following. The algebra of Section 5 and of the polynomial χ_0 in Section 6 is the theory of matrices with equal principal minors [7, 9, 2, 4]. We did not find Theorem 3.1, the role of the start law (the unordered pairs of flows in Theorem 4.1, the second matrix in Proposition 5.1, condition (4), and Theorem 6.1 as a statement about the law of $L(t)$), or Theorems 7.1 and 7.2. Work on aggregate data from Markov chains and on the embedding problem observes something else, namely how many of a population sit in each state at given times. We have not read [7] and [9] themselves. Their statements are taken from [2] and [4], which agree.

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